

MAT 5153 001

Data Analytics

Course Syllabus - Fall 2025



Course Information

Course Description: Prerequisite: MAT 1214, MAT 2233, and MAT 4113 or CS 1063 or CS 1083 or CS 1093, or equivalent, or consent of instructor. This course is intended to develop practical marketable skills in data analytics using rigorous mathematical methodologies via SQL, Python, and Git. The mathematical topics covered involve: Singular value decomposition, single and multiple discriminant analysis, integral transforms with orthogonal and non-orthogonal functions/data and their connection with regression techniques, and nonlinear discriminants through dimensionality augmentation (artificial neural networks). The course covers techniques to characterize the context of a data problem, and addresses issues related to reproducibility of computational results. (Same as MAT 4153. Credit can not be earned for both MAT 4153 and MAT 5153.) This course has Differential Tuition. Course Fee: GS01 \$90.

Credit Hours: 3

Course Modality: Traditional in-person

Meeting Times

Duration: 08/25/2025-12/12/2025

Campus: Main Campus

Location: [MH 2.01.08](#)

Time(s): MW 4:00 PM - 5:15 PM

Learning Goals

At the end of the course, you will be able to:

Analyze, visualize, and interpret complex datasets

Program essential algorithms, instead of using toolboxes, to explore the discrete Fourier transform, generalized regressions, clustering algorithms, and artificial neural networks.

Have a sound mathematical foundations in mathematical algorithms dealing with data

Communicate with Me

Instructor Name: Duy Nguyen Vu Hoang

Department

Mathematics

Office Location

FLN 4.01.44

Student Hours

M 3-3:50pm

Email Address: duynguyenvu.hoang@utsa.edu

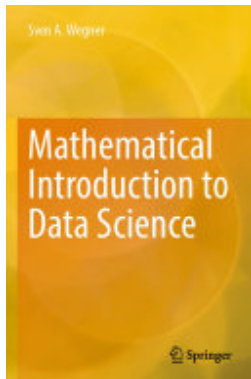
Preferred Method of Communication

email

About Me & My Teaching Philosophy

Successful teaching has no recipe one can follow blindly. I believe that it is important to be flexible, adjust to the students' needs, guide them and walk the extra mile to help them succeed. Any approach I take I keep in mind that **I'm not teaching math, I am teaching students.**

Course Materials



Mathematical Introduction to Data Science

ISBN: 9783662694268

Authors: Sven A. Wegner

Publisher: Springer Nature

Publication Date: 2024-08-30

Assessments and Assignments

Regular assignments will be provided to the student every two weeks.

The student will have two weeks respectively for completion of these problems before the deadline.

The assignments will range from problems needing either a mathematical or computational solution according to the topic of interest (including the latter when/if appropriate).

Activities and Grading

Activity	Quantity	%
Homework	4-5	30
Midterm presentation	1	30
Final term paper	1	40
Total		100%

Distribution of Course Assignments, Their Quantity, and Contribution to Final Grade.

Grade Distribution and Letter Grade

Letter Grade	Grade Range
A+	97.5-100
A	95-97.5
A-	90-95
B+	87.5-90
B	85-87.5
B-	80-85
C+	77.5-80
C	75-77.5
C-	70-75
D+	65-70
D	65-67.5
D-	60-65
F	<60

Grade Distribution and Letter Grade

Guidelines for the Use of Generative Artificial Intelligence

You may not use generative AI to solve homework problems, although a general internet search is allowed and encouraged. You may use generative AI to generate text for the final project, but you must state in a preamble that:

1. I have used generative AI while writing this term paper
2. I have verified all mathematical arguments and checked their correctness.

Course Expectations & Policies

Video and Audio Recording

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded

lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable in the recordings is required before such use unless an exception is allowed by law. For more information on your privacy and class recordings, review [Student Privacy \(FERPA\) in Virtual Classrooms and Other Educational Recordings](#) and the [Guide to Secure Video Conferencing Tools](#).

Course Accessibility Statement

I am dedicated to fostering a learning environment where all students can engage and succeed. If you encounter barriers to your learning, please reach out so we can explore potential design adjustments that may improve access to course materials and activities. In some cases, an approved accommodation through Student Disability Services may be necessary, and I can help connect you with their office if needed. My goal is to collaborate with you to find solutions that maintain the integrity of the course while supporting your learning needs.

Academic Freedom

Academic freedom¹ is a cornerstone of the University. Academic freedom in its teaching aspect is fundamental for the protection of the rights of the teacher in teaching and of the student to freedom in learning.² Each faculty member is entitled to full freedom in the classroom discussing the subject that the faculty member teaches.³ The University of Texas at San Antonio will not penalize or discipline members of the faculty because of their exercise of academic freedom.

Along with this freedom comes responsibility. It is the responsibility of faculty members to ensure that topics taught are related to the classroom subject. Students should be free to take reasoned exception to the data or views offered in any course of study and to reserve judgment about matters of opinion, but they are responsible for learning the content of any course of study for which they are enrolled.⁴ It is not the proper role of the university or any outside agency to attempt to shield individuals from ideas and opinions they find unwelcome, disagreeable, or even deeply offensive.⁵ Engaging with new ideas and perspectives helps students grow intellectually and is beneficial to the educational process.

1. Statement adapted from Texas A&M University's [Syllabus Statement Regarding Academic Freedom](#)
2. 1940 Statement of Principles on Academic Freedom and Tenure
3. Board of Regents Rule 31004; HOP Policy 4.02
4. American Association of University Professors Joint Statement on Rights and Freedoms of Students
5. The Chicago Statement

Syllabus Changes

The syllabus is subject to change at the instructor's discretion. Any changes/corrections to the course materials, assignment dates, or other updates will be communicated to the students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

Course Schedule

For a list of important university-wide dates, review [One Stop's academic calendar](#).

Due Date	Activities/Assignments	Points
9/26/25	HW 1	100
10/17/25	HW 2	100

Department Policies & Resources

Ombudsperson

The ombudsperson is an advocate who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor.

Contact Ilse Rosales Martinez at ilse.rosalesmartinez@utsa.edu

Tutoring Services

UTSA provides many opportunities for student success, including tutoring services for specific math courses. The list below is subject to change based on availability.

MAT 1023, MAT 1043, MAT 1053, and MAT 1073

- We offer peer-to-peer academic support to students enrolled in a variety of courses. Both drop-in tutoring and one-on-one appointments are available.
- **Location:** [Tomás Rivera Center for Academic Excellence](#) MS 2.02.18 (behind the Rowdy statue)

MAT 1053 and MAT 1073

- Synergistic Targeted Academic Review Sessions (STARS), peer-led, interactive, and group focused sessions which cover traditionally difficult topics. The schedule will be provided by instructors through Canvas.
- **Location:** Determined by STARS tutors

MAT 1093, MAT 1213/1214, and MAT 1223/1224

- The Math Gym is unique in that it's a WeBWork lab designed to offer our STEM core students support from math majors and minors through targeted SI sessions. The schedule will be provided by instructors through Canvas
- **Location:** [Math Gym FLN 03.02.08](#)

Note: Course-specific support for upper-division courses may be available for a particular semester. Please request this information from your instructor.

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment

and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.

- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support](#) page.
- UT San Antonio provides reasonable accommodations to students via [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit www.utsa.edu/disability or call (210) 458-4157.
- Your well-being is important. In addition to academic growth, life as a student can include many emotional experiences and opportunities. Being mindful of your own needs and available support can provide you with the tools to successfully navigate these experiences. Student support for well-being, including [24/7 mental health services](#), can be accessed from the [Well-being at UT San Antonio](#) site for students. If needed, we can schedule a meeting, and I can connect you to resources. You are not alone; I am available to help you find resources that might best meet your needs.
- Students at UT San Antonio are responsible for ensuring their work is consistent with UT San Antonio's standards for academic integrity. Students should review [Section 203 of the UT San Antonio Student Code of Conduct](#) for appropriate standards of academic integrity.
- UT San Antonio provides numerous services for students from counseling to tutoring to a food pantry. Visit [Student Affairs Programs and Services](#) and [Student Success](#) for more information.
- Visit the [UT San Antonio Libraries and Museums](#) site for access to journals, research tutorials, and tech gear you can borrow and to find your department's librarian.
- Enroll in the [Roadrunner Success Playbook](#), an open-enrollment, self-paced, online hub in Canvas tailored to ensure you have the resources you need to excel at UT San Antonio.
- Follow [Digital Learning Netiquette](#) standards for your online communication activities.

Student Ombudsperson

Should you have a complaint about this course that cannot be answered by the instructor, or which pertains to unjust or discriminatory treatment, please contact our third-party department representative: Ilse Rosales(ilse.rosalesmartinez@utsa.edu). Ms. Rosales can assist with an investigation and resolution of the matter.

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